

Advanced Blockchain Security — Report

AI-Driven Advanced Blockchain Security Framework using Zero-Trust and Anomaly Detection
Overview: We integrate IsolationForest-based anomaly detection with a zero-trust policy for node operations.

Blockchain Simulator: Toy chain with mining and a security hook that filters suspicious transactions.

ML: IsolationForest trained on synthetic data with features: amount, frequency per hour, gas price, contract_call, hour.

Zero-Trust: JWT-based node tokens required for /tx and /mine endpoints.

Smart Contracts: VulnerableBank (reentrancy) vs SecureWallet (guarded) with mini-audit.

Dashboard: Streamlit app to visualize anomaly scores and top suspicious transactions.

Results: High separability of anomalies; extendable to real chains via web3.py.

Future Work: GNN over address-graph, federated learning, on-chain risk scores.